

Weekly Report Content

General Information

- Group ID: 53
- Group name: Echoes of the Ashen Vault.
- Project name: Echoes of the Ashen Vault (Dungeon Game).
- Date range: 27.07.2026 - 01.08.2026

Tasks completed this week

25125027 - Nguyễn Tấn Minh:

- Re-design the base **Chamber** class to following the Single Responsibility Principle.
- Implement **GauntletChamber**: wave tracker, 0s gap between waves (next wave begins the instant the last enemy of the prior wave dies — no delay), **waves** : **vector<vector<EnemyType>>**, **currentWaveIndex**, and **applyGauntletHeal(Player&)** granting +25% Max HP heal on chamber clear. This heal triggers at the end of Level 1 Ch.3, Level 2 Ch.3, and Level 3 Ch.3.
- Implement **MidChamber**: sets **Player::inMidChamber = true** on **onEnter()** (suspending the 4.0s switch cooldown entirely) and **false** on **onPlayerExit()**. On exit, grant the last-active form a flat +15 Momentum bonus (clamped to 100).
- Reimplement chamber grid logic (support more complicated objects like stairs, etc.).
- Implement level progression flow in **GameState: Level 1 -> Level 2 -> Level 3 -> Boss**. Wire chamber completion → next chamber transition with Mid-Chambers in between.
- Implement chamber retry on death (restart current chamber, preserve prior chamber results per §6.2).
- Create Level 2 and Level 3 chamber layout data files (JSON/CSV) for MapLoader.
- Reimplement logic camera view.
- Create **BossChamber** wrapping **BossMalachar**. **BossChamber** owns the platform layout (6 platforms, radius 3 each for Phase 3; Phase 4 shrinks each at 0.1 units/sec, floor 1.5 units). Phase and platform state are managed here; AI behaviour is delegated to **BossMalachar**.
- Implement **BossMalachar** class with **currentPhase**, **transitionPhase(int)**, **platformSunder(Chamber&)**, **soulLance(Chamber&)** (per architecture.puml).
- Implement Phase 1 behaviour: **Void Bolt Cycle** (3 ranged bolts, Dmg 14 each, fired 0.8s apart, 0.5s charge-glow telegraph — extended to 1.1s total if Foretell is active from a fully-intact Clarity Shard); **Summoning Burst** (spawns 2× Shard Wraith, HP 30, Dmg 8, fly toward Serin). Fixed 12-second repeating cycle. Phase 1→2 transition check at 75% HP (937.5 HP).
- Implement **ShardWraith** enemy (HP 30, Dmg 8) — spawned by BossMalachar Phase 1.
- Implement Phase 2 (HP ≤ 75% → HP ≤ 50%): read **RunState** to conditionally enable Marrow Echo regen (**applyMarrowRegen(float dt)**, 2% MaxHP/sec = 25 HP/sec, active through Phases 2–4) and Obsidian Key blink (**performBlink()**, every 6–9s randomised, active through Phases 2–3). Trigger Resonance Core transition burst on the 75% threshold crossing: **resonanceCoreBurst()** deals 8% of Malachar's current HP (doubled to two sequential bursts ~1s apart if Resonance Core was fully intact).
- Implement Phase 3 (HP ≤ 50% → HP ≤ 25%): notify **BossChamber** to shatter the floor into 6 floating platforms. Add **Platform Sunder** attack every 15 seconds: telegraphs Serin's current platform, which collapses 3 seconds after telegraph (forcing relocation). Carry over Phase 2 modifiers (Marrow regen, Obsidian blink). Trigger second Resonance Core transition burst at the 50% threshold crossing.

- Implement Phase 4 ($HP \leq 25\% \rightarrow HP = 0$): notify **BossChamber** to begin shrinking platforms at 0.1 units/sec (floor radius 1.5). Add **Soul Lance** attack every 10 seconds: single targeted bolt, Dmg 30, 1.0s telegraph (extended to 1.6s if Foretell active). Carry over Marrow regen if active; Obsidian Key blink does **not** carry into Phase 4 (shrinking platforms replace it). Trigger third Resonance Core transition burst at the 25% threshold crossing.
- On phase transition, read **RunState** to apply all relevant modifiers: Hollow Bell stolen $\rightarrow$ activate **reflectWardActive** (first hit every 8s reflects 20% back to Serin, Phase 1 only); Foretell from Clarity Shard collected at any Power $\rightarrow$ begin extended telegraphs in Phase 2 (or Phase 1 if fully intact).
- Boss sprite/animation assets — design and implement multi-phase boss visuals.

25125061 - Châu Tấn Phát:

- Implement **SiegeWraith** (HP 40, Dmg 10; on death calls **explode()** dealing 15 AOE damage in a 3-unit radius — can chain into other nearby enemies including other Siege Wraiths).
- Implement **ChoirHusk** (HP 20 standard / 16 carrier / 22 guard, Dmg 7, Speed 4.0). Call-and-response: one Husk "calls," triggering **triggerCallResponse()** which gives up to 2 nearby Husks a 0.6s shared windup before all strike together. Bonus: killing 2+ Husks within the same 0.6s window grants +1 bonus fragment (tracked in chamber update).
- Implement **ResonantCantor** (HP 50, Dmg 9, does not move). Every 5 seconds emits a pulse applying **SlowedEffect** to Serin if within 6 units. Slow refreshes on re-application rather than stacking in magnitude.
- Implement **HushedStalker** (HP 22, Dmg 9). Invisible until 0.4s attack wind-up telegraph. **onSlowedApplied()** sets **visible = true** for the Slowed effect's duration. Kills while Slowed drop 2 fragments and do not trigger the +1-spawn noise penalty for that specific kill. The noise mechanic (**+1 spawn per Serin offensive action**, hard cap +12 total) is tracked in **ProtectChamber** (Level 3, Chamber 1) and fed to **EnemyFactory**.
- Implement **MirrorBearer** (HP 18 carrier / 24 guard, Speed 6.5). 1 real carrier, 2 decoys (randomised per attempt). Real carrier: **isRealCarrier()** returns true, staggers visibly for 0.5s on any non-lethal hit, drops 3 fragments on death. Decoy: no knockback reaction, no stagger, **isRealCarrier()** returns false, 0 fragments, shatters into smoke on death.
- Implement **VoidShunter** (HP 25, Dmg 11 charge-only, Speed 6.0 when charging).
- Implement basic camera system: follow player position, smooth lerp, clamp to chamber bounds.
- Complete **HUD** as concrete **EchoObserver**: render health bar (sized to active form's MaxHP), active status effect icons, active form indicator, 3 persistent Momentum meters (one per form, frozen meters visually distinct), switch cooldown indicator, and Echo Power bar via **onEchoPowerChanged()**.
- Integrate sound effects (form switching, sword swings, projectile fire, enemy hit, fragment pickup, Echo collection, boss phase transitions).
- Design and implement particle effects: fragment pickup sparkle, Echo collection glow, enemy death poof, boss phase transition bursts.
- Wire **SaveLoadManager** to correctly serialise all **RunState** fields. Verify that loading correctly restores per-form Momentum, Echo outcomes, **special1MomentumThreshold**, and Foretell flags so the run resumes identically.

AI Usage Declaration

Tasks done by AI agents using Gemini model, via Antigravity:

- Wave spawner timers and count checks.

- Invisibility shader/sprite blending code for `HushedStalker`.
- Exploding entity radius checker algorithm (reuse `CollisionSolver::circleIntersect()`).
- Enemy sprite generation prompts for new enemy types.
- HUD layout positioning coordinates and bar rendering.
- Save data serialization structures for `RunState`.
- Boss attack pattern timer/state-machine boilerplate.
- Particle system basic framework.

Issues

No issues encountered during this week.